

# Functional Outcomes Following Critical Illness

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## Introduction

Intensive care units (ICUs) traditionally measure performance using mortality and length of stay. However, survival alone does not capture meaningful patient outcomes.

Without routine measurement of functional recovery, health systems do not know whether the care they deliver leads to an outcome that patients find meaningful.

We developed a registry-based system to collect and report patient-reported functional outcomes (PROMs) after ICU admission across Australia and New Zealand.

## Methods

We are conducting automated text message surveys of ICU survivors from 17 hospitals across Australia and New Zealand at three months post-discharge. Since May 2024, over 17,000 PROM surveys have been sent to patients or their nominated contact.

We used the EQ-5D-5L (assessing mobility, self-care, usual activities, pain, and anxiety) and a 7-point Global Rating of Change scale to compare function at 3 months to self-reported baseline function.

This data is linked to the ANZICS Adult Patient Database registry, which collects information on frailty (clinical frailty scale), diagnosis, and illness severity.

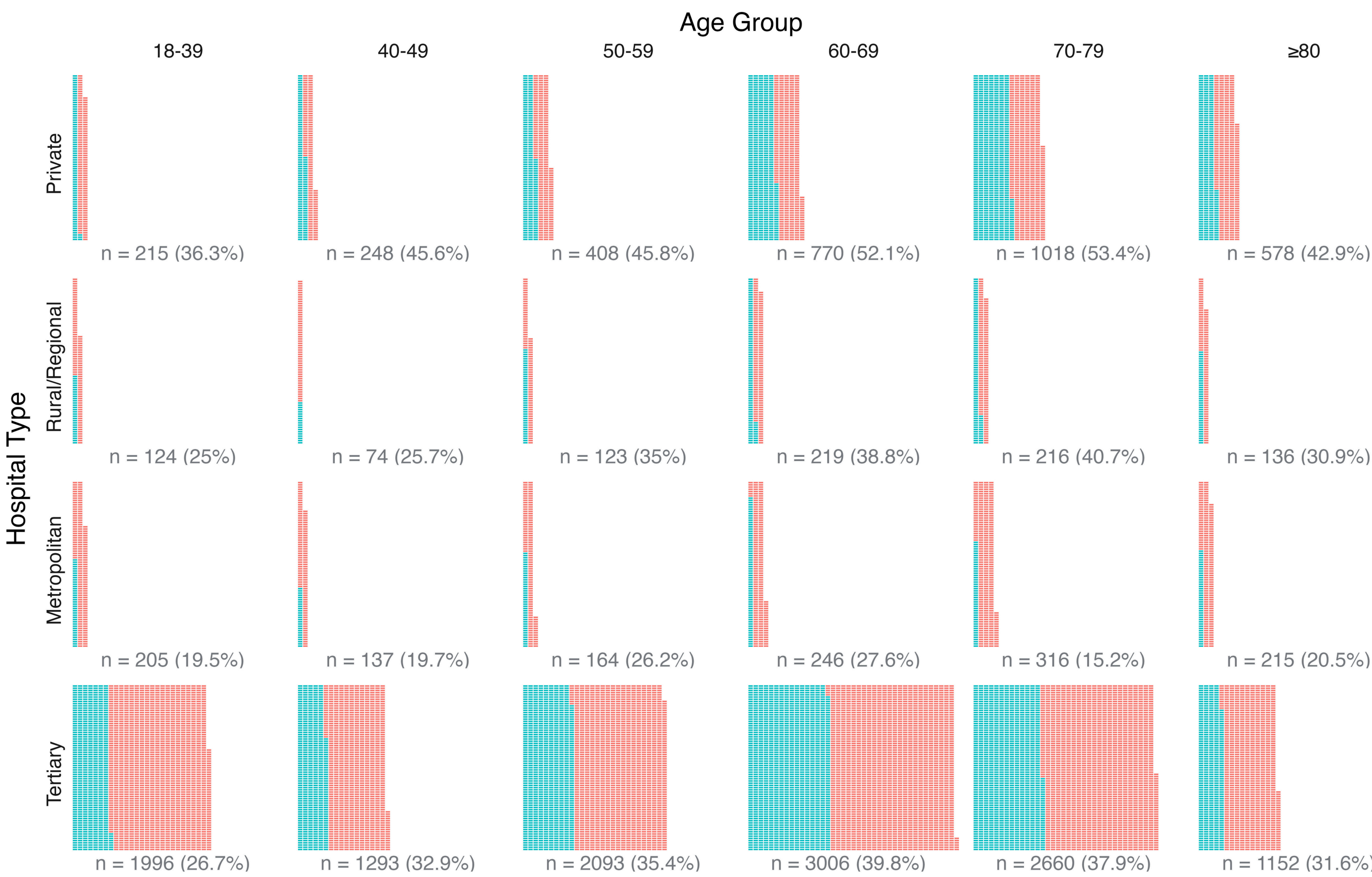
## Results

### PROMS Responses by Age and Hospital Type

Each dot indicates a patient who was sent a survey. **Blue dots** indicate a response, **red dots** indicate no response.

The number of patients in each category, and the percentage of responses is shown below each panel.

Patients under 18 years old are not shown.



### EQ-5D-5L Scores by Age Group

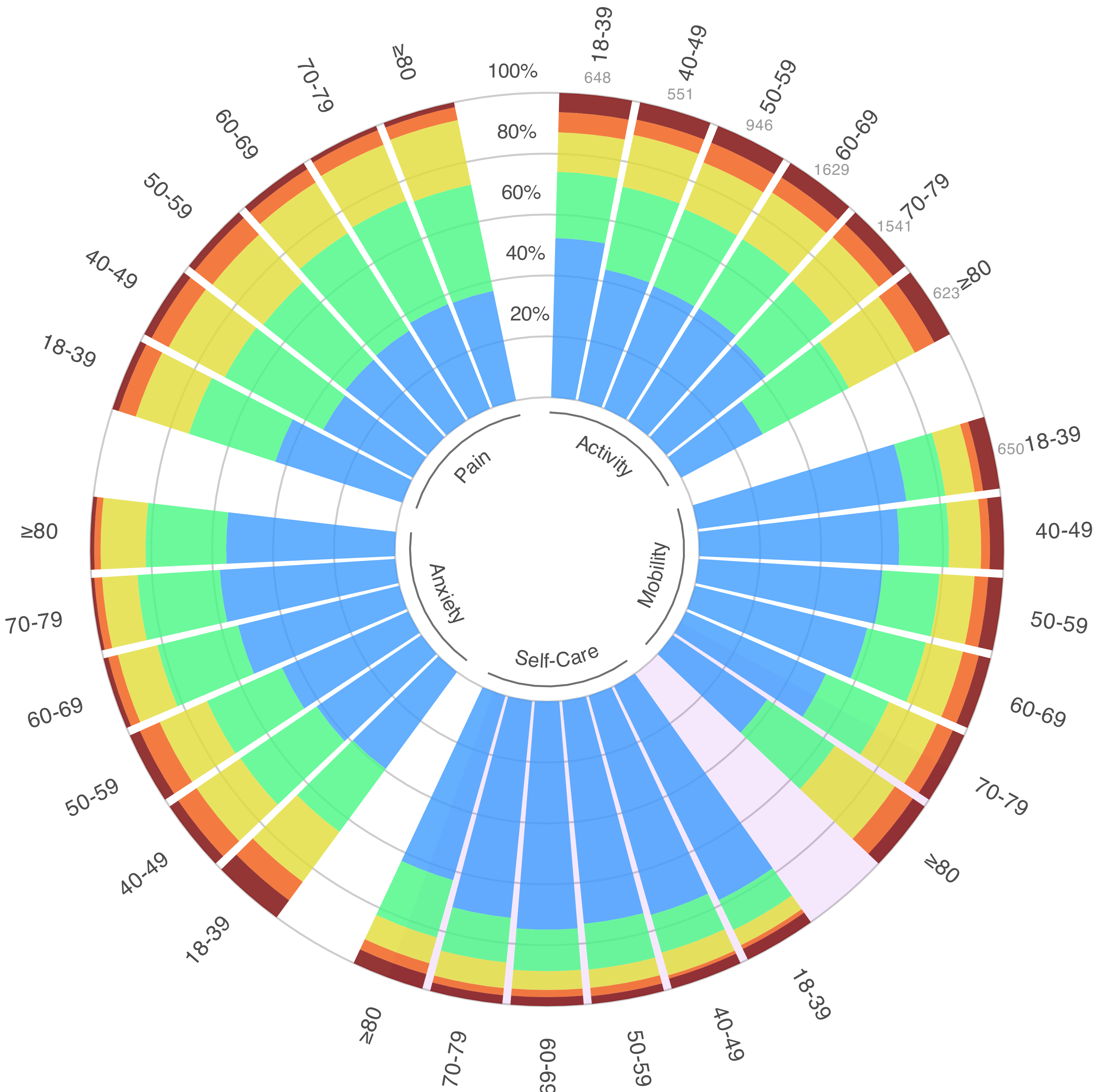
Each column represents the EQ-5D-5L scores for a given age group, with labels in the centre of the figure.

Colours indicate either **no**, **slight**, **moderate**, **severe**, **extreme/unable**.

**Black** column labels above each column indicate age group.

Number of responses for each age group is shown in **grey** above the 'Activity' domain.

Response numbers were identical for each domain.



## Conclusion

Collection of long term functional outcomes from survivors of critical illness is achievable in a registry setting.

**>10% of ICU survivors** report at **least moderate problems** with one or more EQ-5D domains.

The most common problems occur in psychological domains, with **over half** of survivors reporting issues with ongoing pain following critical illness.

## What's Next?

- Benchmark** intensive care unit performance.
- Identify cohorts** with poor functional outcomes.
- Provide data to inform **values-based conversations** with patients about their care.

### Acknowledgements

We would like to thank the following sites for contributing data the PROEMs project: The Alfred Hospital, Alice Springs Hospital, Bendigo Health, Campbelltown Hospital, Canberra Hospital, Fairfield Hospital, Fiona Stanley Hospital, Gold Coast University Hospital, Liverpool Hospital, Nepean Hospital, north Shore Private Hospital, St. John of God Health Care, Sunshine Hospital, Townsville University Hospital, University Hospital Geelong, Wellington Hospital.

